DeepCXR: A Deep Learning Framework for Automated Chest X-Ray Analysis Using ResNet Architectures

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***Abstract*—Chest X-ray (CXR) images play a crucial step in detecting pulmonary diseases and abnormalities in human lungs, and while the medical field evolves rapidly, it remains prone to human error and misjudgment [1], deep learning models have been widely applied to this field in the past years, however many deep architectures suffer from the vanishing gradients problem, to address this the proposed model utilizes the ResNet18 architecture, employing skip connections and residual blocks to allow for the training of deeper layers without a loss in accuracy, by implementing transfer learning from models pre-trained on large-scale datasets [2], the model efficiently extracts complex radiographic features, such as lung opacities and anatomical irregularities, results show that the ResNet approach helps the model catch details that can be often missed by human eye or basic models, achieving an image level AUROC of 0.95 and an F1-score of 0.90. DeepCXR acts as a reliable assistant for radiologists. To enhance diagnostic consistency, the system incorporates a memory bank that stores and retrieves historical feature representations of radiographs. This mechanism allows the model to perform large-scale comparisons across diverse samples, improving its ability to identify rare and subtle anomalies, demonstrating the potential of DeepCXR as a reliable and scalable computer-aided diagnosis system for chest X-ray analysis.**

Keywords—Chest X-ray, Deep Learning, Medical Image Analysis, Memory Bank, Pneumonia Detection, ResNet-18, Transfer Learning.

I. Introduction

Chest X-rays (CXRs) are the most popular type of x-ray globally, used for screening and identifying many types of lung diseases, including Pneumonia, Covid-19, Tuberculosis, and many more [3]. Since so many of these diseases are diagnosed by CXR, radiologists can be under a lot of pressure to read many screenings daily and correctly identify them, preventing diseases from reaching a lethal state and dealing with them as early as possible.

Advanced models and algorithms have been developed over the years to assist radiologists detect lung diseases, and make them more efficient at identifying healthy or unhealthy lungs. Trained tools emerged as technology grew, with technologies becoming more advanced as time moved on.

Early detection methods used traditional ways, such as local texture analysis combined with K nearest neighbors, to detect lung abnormalities [4] There are many traditional classification algorithms, such as Support Vector Machine (SVM), which is a strong mathematical model for classification and regression [5]. Other traditional machine learning methods have also seen use, some of which being ensembling methods, such as Random Forest, which are strong contenders for classification tasks [6]. As deep learning grew, new ways emerged of classifying CXRs, most notably, Convolutional Neural Networks (CNNs). Using supervised training methods, such as binary classification, some researchers were able to achieve significantly better results, and improve these tools even more [7]. However, these tools are not without limitations. Supervised classification techniques rely on quality labeled data, and can be very sensitive to noise, such as improper labeling. They also struggle on rare diseases, as these diseases do not have a lot of data available, which impacts the accuracy of the model, and could even create a sense of false competence [8].

These limitations, with the addition of the cost of data labeling, have caused researchers to shift to semi-supervised techniques. This type of training trains the model on healthy lung images, which causes it to identify any unhealthy lungs, since the model was trained on healthy lung images only, it can detect any defection without explicit training on that type of defection. Transfer learning is another method that was utilized, however, most pretrained models are hard to adapt to CXR due to low transfer efficiency because of the dissimilarity of natural images and CXR. This can cause difficulties in training models because of the lack of quality CXR data available, where models must be trained from scratch or created by fine tuning on a model already pre-trained on CXR data, which might not be of high quality because of the limitations mentioned earlier.

Interpretability is another major area of focus in CXR detection models, as radiologists cannot blindly rely on a prediction made by a model. Detection models must have some way of explaining their decision. Localization makes this happen by drawing heatmaps over abnormal areas, therefore explaining exactly why it made that decision.

While current models perform well during development, they can sometimes suffer from significant domain shift when exposed to images from different hospitals, which can cause them to perform poorly in production. As mentioned earlier, the lack of data quality is also a problem, training models on a rare disease or an imbalanced dataset can backfire, creating a model that achieves high accuracy, but does not actually perform well [2].

This paper attempts to solve these limitations through DeepCXR, a model trained only on healthy images, with ResNet18 [11] as the backbone, this model can recognize unhealthy lungs without prior training, and can generalize to different hospital scanners with only a few images. This model also provides feature explainability, as it creates heatmaps over the images to highlight exactly where the problem is. The model is also lightweight due to the small coreset subsampling ratio.

II. Related Work

Lee et al. [12] presented a deep Learning in Medical Anomaly Detection Early methods for detecting diseases in medical images mostly used supervised learning technique, which requires labeling thousands of images, this approach will take time and can be prone to error, to solve this issue recent research focused on unsupervised anomaly detection, which trains the model only on normal (healthy) images, then testing the model to see if it's able to classify normal and abnormal chest X-ray images.

Newer methods like PatchCore improve accuracy by extracting small patches of features from healthy images then storing them in a memory bank. When testing a new image, the model compares the image features to stored healthy ones. This method effectively preserves details that might get lost in other models due to the vanishing gradient problem.

Comparison with State-of-the-Art (2024) A recent study [12] titled "Region and Global-Specific PatchCore based Anomaly Detection from Chest X-ray Images" applied this exact PatchCore architecture to chest X-rays, their method added complex preprocessing steps, including affine transformations to align specific organs, arguing that perfect alignment was necessary for accuracy. However, despite these computationally intensive additions, their approach achieved a maximum AUROC of 0.774

Our Approach: in contrast, this project demonstrates that such complex alignment steps are unnecessary and perhaps detrimental to the model. The project implemented a streamlined PatchCore model using coreset sampling (0.005 ratio) to create a highly efficient memory bank without the pre-alignment step, By focusing on the most representative 0.5 percent of healthy features, our model achieved a significantly higher image AUROC of 0.95 and an F1-Score of 0.90, These results confirms that using a simpler method and optimized memory bank approach not only reduces the computational load, but can also outperform complex alignment-based pipelines in detecting pulmonary abnormalities.

J. Sato et al. [13] presented a supervised deep learning methods require a huge amount of labeled images, which can be very costly or might cause problems unseen until real world use, such as noisy data.

This has caused the shift to unsupervised methods, which do not suffer from the lack of quality disease data, as they are only trained on healthy images, but they are not without limitations as well.

This caused the researchers of this paper to come up with AnatPaste, which is an anatomy-aware self-supervised learning (SSL) based augmentation technique, that can create anomalies in healthy lung images, eliminating the need for labeled disease data, as the model can create realistic looking augmentations, which helps the model recognize what an unhealthy lung looks like.

Their model was able to achieve an AUC of 92.1%, 78.7%, 81.9% on different datasets, which at the time, was higher than any existing UAD models. In comparison, the approach presented in this paper was able to achieve an AUC of 95%, which shows a 3% improvement without relying on expensive augmentation techniques [13].

The model presented in this paper was able to achieve significantly better results through utilizing PatchCore [14], using ResNet18 as the backbone to extract high quality patches into a memory bank, which then can be sampled using Coreset Sampling [15] with a 0.5% ratio to achieve a better performance than other techniques, without using any computationally expensive methods. The simple approach used in this paper has also achieved an F1-Score of 90%, which is a testament to its capability.

Nguyen et al. [16] presented one of the biggest problems in medical AI is that the model works perfectly during development, but when tested on real hospitals, it performs poorly. This happens due to the fact that hospitals use different X-ray machines, different patients, and different lighting, a major study by Nguyen et al. (2022) [16] tested this exact problem, their proposed solution was installing a large AI system at Phu Tho General Hospital in Vietnam to help their doctors to detect common lung diseases like Tuberculosis and COVID-19.

What they did: they used a supervised machine learning model and trained it on thousands of labeled images (6,285) of real patient X-rays from the hospital's daily work, the results were not as impressive as the idea sounds, the system achieved an F1-score of 0.653 on real life data from other hospitals, indicating that the model struggled in the real world data from other locations.

The reason as the authors explained was that their model was confused because the hospital's X-rays looked slightly different from the training data, but the model was trained to look for a specific type of disease. This slight difference confused it, indicating that the model did not generalize enough to this task. On the other hand, the proposed DeepCXR model was only trained on normal lung images from three different datasets, instead of looking for the abnormalities it looks for what a normal lung looks like using the memory bank method (PatchCore).

The Comparison: the supervised system by Nguyen et al. struggled with an F1-Score of 0.653, In contrast, our unsupervised model achieved a much higher F1-Score of 0.90.

The Conclusion: this proves that learning normality is a safer and more reliable way to screen patients than trying to memorize every possible disease out there.

Pillai, A.S. [17] proposed a supervised deep learning model, using DenseNet121 trained on a downsampled version of Chexpert data, which contains 13 classes of different lung diseases. Their best performing model achieved an AUCROC of 77%, and an F1-Score of 27%. One of the limitations mentioned

in their paper was the huge size of the dataset, which was 419GB. They attempted to solve this by downsampling the dataset, bringing its size down to 11GB. Their model performed very inconsistently, scoring an AUCROC as high as 96% on "Pleural Other", but at the same time scoring as low as 22% on "Lung Lesion". The authors believe that this was due to the imbalance of the dataset, which significantly reinforces the need for an approach such as DeepCXR. DeepCXR does not suffer from any of the problems faced in that study, since it only trains on healthy images, which reduces the training time and the need for quality labeled disease datasets, which have proven to be difficult to obtain. In comparison with the model proposed in Pillai's paper, DeepCXR achieved better results, scoring an AUROC of 95%, and an F1-Score of 90%, compared to their model which achieved an AUROC of 78%, and an F1-Score of 27%.

III. Methodology

A. Framework Overview

The DeepCXR framework is an unsupervised anomaly detection designed for the chest radiographs tasks, Unlike traditional deep learning that is based on supervised learning technique that require thousands of hand-labeled images for every specific disease, DeepCXR on the other hand only needs the normal chest unlabeled images "One Class Classification".

The architecture is built upon the PatchCore methodology, which focuses on mid-level feature representations, by using ResNet-18 as a backbone, the framework extracts high-dimensional patches from the input image. To ensure that the system remains computationally efficient for clinical use which might not have advanced computers infrastructure, DeepCXR uses a coreset-sampled memory bank, this "bank" serves as a reference library of health lung features [18] by using a greedy sub-sampling ratio of 0.005, the framework keeps only the most mathematically diverse 0.5% of normal features that the model was early trained on, this compression allows the model to perform lighting fast comparison during the diagnostic phase.

When a new patient scan (chest X-ray) is introduced, the framework (model) performs a Nearest Neighbor Search against this memory bank. If a patient's scan differs significantly from the stored 'memories' of healthy tissue, the system flags the area with a high anomaly score and highlights the suspicious region, this approach allows DeepCXR to achieve high sensitivity in detecting subtle pathologies—such as early stage pneumonia or tuberculosis that might be overlooked during the standard manual inspection (Fig. 1).

TABLE I. Comparison of Related Works

Research	Comparison		
	Methodology	**Key Limitations**	**AUROC**
Lee et al.	PatchCore	Computationally intensive due to complex organ alignment	0.774
J. Sato et al.	AnatPaste	Relies on synthetic augmentation (fake anomalies)	0.921
Nguyen et al.	Supervised Learning	Had trouble generalizing to real data	-
Pillai, A.S.	Supervised Learning	Class imbalance	0.77

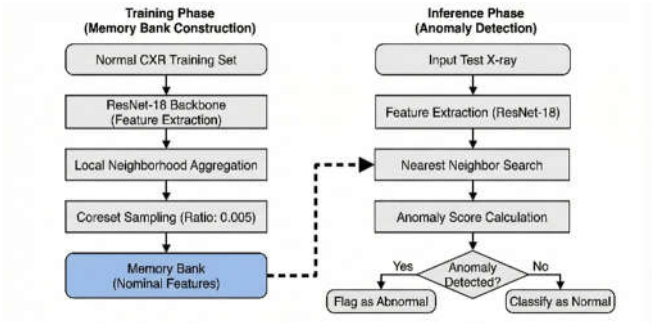

Fig. 1. Flowchart of the DeepCXR Methodology

B. Dataset Description

To ensure generalization of the DeepCXR framework, a multi-source data strategy was used; by combining images from three different medical datasets, the model was then trained on a diverse range of patient demographics, imaging hardware, and pathological variations.

1) Data Sources: The datasets consists of the following primary sources:

- Tuberculosis (TB) Chest X-ray Database: A collaborative database from Qatar University and the University of Dhaka, providing 3,500 Tuberculosis-positive cases and 3,500 normal images [19].
- Pediatric Pneumonia Dataset: sourced from the Guangzhou Women and Children's Medical Center, containing 5,863 pediatric X-ray images. This dataset is noted for its high diagnostic quality, verified by a three-physician expert panel [20].
- CheXImageNet Dataset: A balanced collection contributing 1,626 COVID-19, 1,802 Normal, and 1,800 Pneumonia images, sourced from international repositories including Eurorad and Radiopaedia [21].

2) Data Integration and Preperation: Instead of using just one dataset, this project combines images from multiple sources into a single dataset, this ensures the model learns what a 'normal' lung looks like from a wide variety of healthy examples, making it able to correctly classify wide range of lungs X-ray images from different sources.

3) Training Set (Nominal Baseline): All "Normal" labeled images from the three datasets were put into one training dataset. This provided the ResNet-18 backbone with a massive and diverse library of healthy lung images, covering different age groups and acquisition conditions.

4) Test Set (Pathology Distribution): The abnormal images were categorized by pathology (Tuberculosis, Pneumonia, and COVID-19) for the evaluation phase, where 300 images from each dataset were taken and placed in a single testing dataset. Which then resulted on achieving a high AUROC and F1-Score as the paper mentioned earlier.

While medical data is often imbalanced (having more healthy images than sick ones) DeepCXR framework handles this perfectly because it is unsupervised (it trains only on healthy images), making it impossible for the model to be biased by a

lack of disease samples, to strictly prove its performance, the model was evaluated using the AUROC metric (achieving 0.95) rather than simple accuracy, this metric mathematically confirms that the proposed model correctly distinguishes between healthy and abnormal cases regardless of the ratio of sick patients in the test set.

C. Feature Extraction

To tackle the problem of CXR anomaly detection, this project used PatchCore, which is a memory bank algorithm. This method assumes that anomalies exist as local structure deviations from the regular healthy structure.

This strategy follows 3 main stages: patch extraction, coreset subsampling, anomaly scoring.

1) Patch Extraction: ResNet18 [11] was used as the backbone for feature extraction, which is a pretrained classification model. PatchCore extracts the feature from the 3rd layer, then maps them into a feature space [14]. Average global pooling is used to ensure the extracted embeddings are robust to slight misalignments.

2) Coreset Subsampling: Storing every embedding extracted would be extremely expensive computationally. This is why PatchCore employs greedy coreset subsampling. This approach selects a representative subset of the data, to make sure most of the information is captured. This is achieved by selecting a set of points that maximize the distance to the current sets of points.

3) Anomaly Scoring: For inference, the anomaly score for a CXR image is determined by the distance between its embeddings, and the closest neighbors in the memory bank. For each patch in the test image $\mathcal{P}(x)$, the anomaly score $\mathcal{S}(x)$ is calculated using Euclidean distance, from the patch to its nearest neighbor in the coreset subsample $\mathcal{M}$ (1).

$$\mathcal{S}(x) = \max_{p \in \mathcal{P}(x)} \left(\min_{m \in \mathcal{M}} \|p - m\|_2 \right) \tag{1}$$

Identifying re-weighting is applied to improve robustness against outliers (Fig 2).

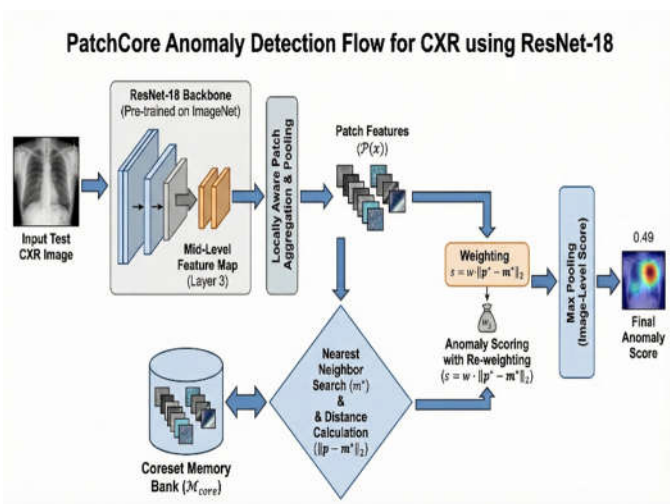

Fig. 2. A Flowchart of Patchcore's Workflow

IV. Results & Discussion

A. Evaluation Metrics

To accurately assess the performance of DeepCXR, two of the most used metrics in unsupervised classification and medical imaging were used: Area Under the Receiver Operating Characteristic (AUROC), and F1-Score.

1) AUROC: AUROC was considered the primary metric in this project, due to its robustness, and its focus on true positives and false positives.

The AUROC works by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at different thresholds. True Positive Rate (TPR) is given in (2) and False Positive Rate (FPR) in (3).

$$\text{TPR} = \frac{\text{TP}}{\text{TP+FN}} \tag{2}$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP+TN}} \tag{3}$$

Where TP, TN, FP, and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively. The AUROC is then calculated as in (4).

$$\text{AUROC} = \int_0^1 \text{TPR}(x)dx, \text{ where } x = \text{FPR} \tag{4}$$

In CXR anomaly detection, having a higher AUROC means the model performs better.

2) F1-Score: AUROC measures ranking quality, however, in a real world environment there has to be a definite threshold that is used to get an explicit result, either abnormal or healthy. F1-Score measures the model performance at a specific threshold, which is different than how AUROC does it.

F1-Score is measured using precision and recall, as given in (5) and (6).

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}} \tag{5}$$

Recall is the same as TPR mentioned earlier. Combining precision and recall, we get the following equation for F1-Score.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{6}$$

Maintaining a high F1-Score ensures that the model is not only performing well during development, but will also perform well in real world use.

In CXR anomaly detection, some of the notable metrics include F1-Score, and Receiver Operating Characteristic - Area Under the Curve (ROC-AUC), which are the usual metrics used for anomaly detection, since accuracy can be misleading due to the imbalance of classes. ROC-AUC ignores the count of negatives, instead, it calculates the True Positive Rate (TPR) versus the False Positive Rate (FPR), which makes it significantly more robust. The same idea can be said for the F1-

Score, however F1-Score measures at a specific threshold, whereas ROC-AUC measures at every possible threshold, which allows it to achieve a higher percentage.

B. *First Experiment*

The first experiment in this project utilized a Coreset Sampling ratio of 1%, with ResNet18 [11] as the backbone. In this experiment, a single dataset was used, which contains 2 classes, healthy and pneumonia [22]. However, this approach proved to be not feasible due to the large computational cost, as it required a significant amount of Virtual Random Access Memory (VRAM).

C. *Second Experiment*

In the second experiment, multiple different datasets were used to test different diseases, and to simulate images from different scanners and hospitals. A coreset sampling ratio of 0.5% was used, with ResNet18 [11] as the backbone once again. This approach significantly reduced the computational power required, and provided the following results:

- ROC-AUC of 95%.
- F1-Score of 90%.

These results prove the model's ability to generalize to unseen data and perform on images of different scanners and hospitals.The images in

Fig. 3 showcase what defect lungs (pneumonia) look like. These images can be very time consuming for radiologists to detect any defections, as they are not that obvious. This is why DeepCXR creates heatmaps to help radiologists detect exactly which areas look abnormal and circles them. As can be seen in Fig. 4, the model created an anomaly map of the image, showing exactly where the defection is and circles it.

Fig. 5 shows how the model does not circle anything in the healthy image. This makes it clear that the model believes nothing looks abnormal about this image.

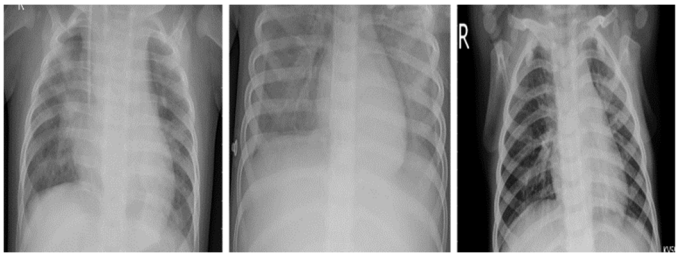

Fig. 3. Images of Defect Lungs Sampled Randomly From the Dataset

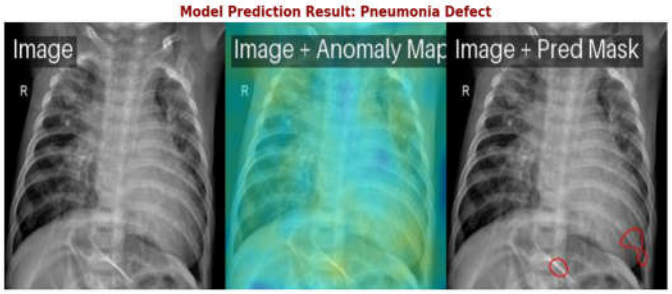

Fig. 4. Model Result on a Random Defect Image

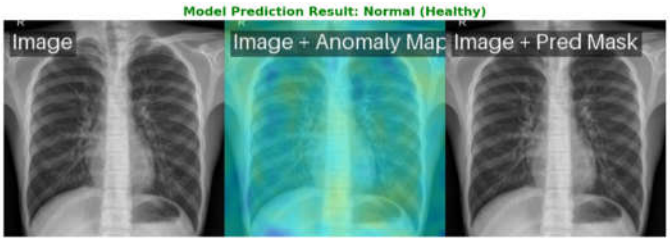

Fig. 5. Model Result on a Random Healthy Image

DeepCXR shows that you do not have to sacrifice interpretability to achieve good results, as having models that are explainable is extremely important in medical fields. This project shows that it is possible to achieve great results without having to resort to the black box of artificial intelligence.

DeepCXR was able to solve many of the problems encountered in similar works, without sacrificing performance or accuracy. The proposed model does not require labeled datasets of diseases, as opposed to models that used supervised learning, which makes it less susceptible to the lack of quality data. DeepCXR is also a performant model, which means it is quick to make predictions and does not require a significant amount of preprocessing of images.

V. Conclusion

In this study, we developed DeepCXR, an unsupervised anomaly detection for chest diseases to assist radiologists in the interpretation of chest radiographs, using a ResNet-18 backbone combined with a coreset-subsampled memory bank, this method delivered a model capable of detecting diseases and abnormalities in human lungs in an accurate, fast and reliable way.

The results shows that DeepCXR model achieves an image level AUROC of 0.95 and an F1-score of 0.90, which significantly outperform older methods and traditional ways, this is due to using the coreset sampling method (0.005 ratio), avoiding using complex image preprocessing, and having a generalized dataset from many resources and different X-ray imaging techniques, all of this can deliver a model capable of achieving high accuracy with the least computational resource possible. Ultimately, DeepCXR offers a reliable and scalable solution for clinical environments and hospitals, its ability of remembering and retrieving features allows for accurate detection of subtle pulmonary abnormalities without the need for extensive annotated datasets and complex preprocessing.

TABLE II. Proposed Model vs Related Works

Research	Comparison		
	Methodology	**F1-Score**	**AUROC**
Lee et al.	PatchCore	0.772	0.774
J. Sato et al.	AnatPaste	0.90	0.921
Nguyen et al.	Supervised Learning	0.653	-
Pillai, A.S.	Supervised Learning	0.27	0.77
Our Model	PatchCore	0.90	0.95

REFERENCES

[1] M. Gambato, N. Scotti, G. Borsari, J. Z. Bertoja, J.-D. Gabrieli, A. De Cassai, G. Cester, P. Navalesi, E. Quaia, and F. Causin, "Chest xray interpretation: Detecting devices and device-related complications," Diagnostics, vol. 13, no. 4, p. 599, 2023.

[2] S. Kotsiantis, D. Kanellopoulos, P. Pintelas, et al., "Handling imbalanced datasets: A review," GESTS international transactions on computer science and engineering, vol. 30, no. 1, pp. 25-36, 2006.

[3] X. Wang, Y. Peng, Z. Lu, M. Bagheri, R. M. Summers, and L. Lu, "Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 3462-3471, July 2017.

[4] B. Van Ginneken, S. Katsuragawa, B. M. ter Haar Romeny, K. Doi, and M. A. Viergever, "Automatic detection of abnormalities in chest radiographs using local texture analysis," IEEE transactions on medical imaging, vol. 21, no. 2, pp. 139-149, 2002.

[5] M. A. Hearst, S. T. Dumais, E. Osuna, J. Platt, and B. Scholkopf, "Support vector machines," IEEE Intelligent Systems and their applications, vol. 13, no. 4, pp. 18-28, 1998.

[6] L. Breiman, "Random forests," Machine learning, vol. 45, no. 1, pp. 532, 2001.

[7] M. T. Islam, M. A. Aowal, A. T. Minhaz, and K. Ashraf, "Abnormality detection and localization in chest x-rays using deep convolutional neural networks," arXiv preprint arXiv:1705.09850, 2017.

[8] J. A. Richards, "Supervised classification techniques," in Remote sensing digital image analysis, pp. 263-367, Springer, 2022.

[9] A. Spahr, B. Bozorgtabar, and J.-P. Thiran, "Self-taught semi-supervised anomaly detection on upper limb x-rays," in 2021 IEEE 18th international symposium on biomedical imaging (ISBI), pp. 1632-1636, IEEE, 2021.

[10] S. Xu, H. Wu, and R. Bie, "Cxnet-m1: anomaly detection on chest x-rays with image-based deep learning," IEEE Access, vol. 7, pp. 4466-4477, 2018.

[11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 770-778, 2016.

[12] J. Lee, H. Kim, and S. Park, "Region and global-specific patchcore based anomaly detection from chest x-ray images," Journal of Korea Multimedia Society, vol. 27, no. 2, pp. 25-34, 2024. J. Sato, Y. Suzuki, T. Wataya, D. Nishigaki, K. Kita, K.

[13] Yamagata, N. Tomiyama, and S. Kido, "Anatomy-aware self-supervised learning for anomaly detection in chest radiographs," IScience, vol. 26, no. 7, 2023.

[14] K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, and P. Gehler, "Towards total recall in industrial anomaly detection," in Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 14318-14328, 2022.

[15] A. Dasgupta, P. Drineas, B. Harb, R. Kumar, and M. W. Mahoney, "Sampling algorithms and coresets for \ell_p regression," SIAM Journal on Computing, vol. 38, no. 5, pp. 2060-2078, 2009.

[16] H. Q. Nguyen, K. Lam, L. T. Le, et al., "Deployment and validation of an ai system for detecting abnormal chest radiographs in clinical settings," Frontiers in Digital Health, vol. 4, p. 890759, 2022.

[17] A. S. Pillai, "Multi-label chest x-ray classification via deep learning," arXiv preprint arXiv:2211.14929, 2022.

[18] A. Bulat, E. Sánchez-Lozano, and G. Tzimiropoulos, "Improving memory banks for unsupervised learning with large mini-batch, consistency and hard negative mining," 2021.

[19] T. Rahman et al., "Reliable tuberculosis detection using chest x-ray with deep learning, segmentation and visualization," IEEE Access, vol. 8, pp. 191586-191601, 2020.

[20] D. S. Kermany et al., "Identifying medical diagnoses and guide treatment by a deep-learning-based heterogeneous data," Cell, vol. 172, no. 5, pp. 1122-1131, 2018.

[21] S. Shastri et al., "Cheximagenet: a novel architecture for accurate classification of covid-19," Health Technol., vol. 12, pp. 193-204, 2022.

[22] D. S. Kermany, M. Goldbaum, W. Cai, C. C. Valentim, H. Liang, S. L. Baxter, A. McKeown, G. Yang, X. Wu, F. Yan, et al., "Identifying medical diagnoses and treatable diseases by image-based deep learning," cell, vol. 172, no. 5, pp. 1122-1131, 2018.